SCIENCE HISTORY INSTITUTE

JENNY PICKWORTH GLUSKER

Transcript of an Interview
Conducted by

Jacqueline Boytim and Sarah Schneider

at

Interviewee's home
Huntingdon Valley, Pennsylvania

on

1, 3, and 4 November 2022

(With Subsequent Corrections and Additions)

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> Jenny Pickworth Glusker, interview by Jacqueline Boytim and Sarah Schneider at Interviewee's home, Huntingdon Valley, Pennsylvania, 1, 3, and 4 November 2022 (Philadelphia: Science History Institute, Oral History Transcript # 1142).

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JENNY PICKWORTH GLUSKER

1931	Born in Birmingham, England, on 28 June

Education

1953	BA, Oxford University, Chemistry (first class honors)
1957	MA, Oxford University, Chemistry
1957	DPhil, Oxford University, Chemistry

Professional Experience

	California Institute of Technology
1955-1956	Postdoctoral Research Fellow
	The Institute for Cancer Research
1956	Research Fellow
1957-1967	Research Associate
1967	Assistant Member
1967-1979	Associate Member
1979-2003	Senior Member
2003-Present	Senior Member Emerita (now Professor Emerita)
	University of Pennsylvania
1980-2012	Adjunct Professor of Biochemistry and Biophysics
	Earlier affiliated appointments beginning in 1969

Honors

1968	Rosa Briegel Barton Lectureship, University of Oklahoma
1978	Philadelphia Section Award, American Chemical Society
1979	Garvan Medal, American Chemical Society
1983	Philips Lectureship, Haverford College
1984	Who's Who in Technology
1985	Honorary DSc, College of Wooster
1991	Public Service Award, American Crystallographic Association
1994-1995	Visiting Fellowship, Oriel College, Oxford

1995	Hassel Memorial Lecture, Oslo, Norway
1995	Fankuchen Award, American Crystallographic Association
1997	Visiting Professorship, International Union of Crystallography Suez Canal University
1998-Present	Who's Who in America, of American Women, in the East, in Science and Engineering
1999	Fellow of the American Association for the Advancement of Science
2001-Present	Elected Honorary Fellow of Somerville College (lifetime)
2002-Present	Who's Who in the World
2006	Visiting Professorship, International Union of Crystallography Kayseri University, Turkey Invited lecture, Second Turkish Crystallographic Meeting, Erciyes University, Kayseri, Turkey
2011	Inaugural Fellow, American Crystallographic Association
2011	John Scott Award
2014	Fellow of the Royal Society of Chemistry (FRSC)
2014	International Year of Crystallography (IYCr2014) Opening Ceremony 20 January 2014, UNESCO, Paris, France Lecture: "Crystallography: past, present and future"
2014	Sigma Xi William Procter Prize for Scientific Achievement

ABSTRACT

Jenny Pickworth Glusker grew up in Birmingham, England. Her parents were physicians who had met through their work at a psychiatric hospital, and Glusker had a sister and a brother. She grew up attending a local Presbyterian church. When she was young, her father took her to see King George VI and Queen Elizabeth at an opening celebration of the Queen Elizabeth Hospital.

Many of Glusker's childhood memories are connected to World War II. She would often spend the night in the air raid shelter that her father had set up to protect the family from bombs. When the war ended, Glusker remembers an illuminated bus and street parties that were in Birmingham to celebrate the end of the war in Europe. Though her ability to travel was limited during the war, after the war, she traveled with a school group to Sweden. She also, after her undergraduate studies, took a trip to Morocco.

In Glusker's early school days, there was an emphasis on math and spelling, and she learned practical skills such as cooking. She would have liked to do woodworking, but that was something that girls did not do in her school. When Glusker took her eleven-plus exam to place into high school, she did well and earned a scholarship to attend the King Edward VI High School for Girls. She loved chemistry and would read her mother's textbook on incompatibles and experiment with chemicals that her father gave her. She was encouraged by her chemistry teacher, Yvonne Way, and had hands-on experience carrying out chemistry experiments in class. Glusker also studied Latin, German, and French in school.

Glusker decided to apply to university a year early. She took the necessary exams and interviewed for Somerville College of the University of Oxford with Dorothy Hodgkin. The unknown during the practical exam was sodium thiosulfate, which Glusker was familiar with from working in her father's dark room. She demonstrated her familiarity with using scientific equipment during the exam and was admitted to Somerville College.

At the University of Oxford, Glusker says she was one of just five women in a group of over one hundred people studying chemistry. She had tutorials with Dorothy Hodgkin, who provided feedback on Glusker's essays. Glusker describes recreational activities that students participated in, including sailing and going to dances, and she attended political lectures on campus. Glusker conducted undergraduate research in Harold Warris Thompson's lab. She researched infrared spectroscopy and deuterium chloride, which resulted in published papers. She was the first woman in Thompson's lab and made friends with peers in the lab, including her future husband, Donald L. Glusker.

For her graduate studies, Glusker chose not to continue working with Thompson and instead went to work in Dorothy Hodgkin's lab to study crystallography. When Hodgkin received small crystals of a derivative of vitamin B_{12}, Glusker started working on the structure. Kenneth Trueblood at the University of California, Los Angeles assisted by using newly developed computer technology. Glusker would also use computer card-reading machines in the Mathematical Institute late at night and employed diffraction maps and calculations in her analysis. Eventually, the structure of vitamin B_{12} was identified. Glusker gave the first talk about the findings from Hodgkin's lab, and Hodgkin was excited to have the discovery presented. When Alexander R. Todd would lecture about vitamin B_{12}, Hodgkin would follow him to his lectures and explain to the audience how her lab had identified the structure.

Glusker went to Caltech to do a postdoctoral fellowship, and by that time was engaged to her future husband, Don, who was already working at Caltech. They had a small wedding in California. At Caltech, Glusker worked in the lab of Robert B. Corey and conducted research on peptides.

After one year at Caltech, Glusker moved with her husband to Philadelphia. She began working for Lindo Patterson at the Institute for Cancer Research, which later became Fox Chase Cancer Center. Initially, Patterson did not have enough money for her position, so she agreed to start her work as a technician. After a year, her role and title changed to be more reflective of her expertise. Glusker used crystallographic methods to research the Krebs cycle. She also conducted research on hydrocarbons that go into DNA, enzymes, and neutron diffraction. She had some researchers from abroad work in her lab. Also at the institution were award-winning scientists, including one of the scientists, David Hungerford, who discovered the Philadelphia chromosome. After Lindo Patterson's death, Glusker took over running the lab.

In addition to her work at the Institute for Cancer Research, Glusker was an adjunct professor at the University of Pennsylvania. She wrote a book with Kenneth Trueblood on crystal structure analysis that was translated into Russian and other languages. Glusker also stayed busy serving as a journal editor, volunteering in a local group enhancing science education, and traveling to set up schools around the world on crystallographic teaching.

Glusker explains how she approached work-life balance as she raised her three children and describes what the process was like for her to become a United States citizen. She talks about mentoring other women in the field and describes the influence of William Henry Bragg on inclusion of women in the field of crystallography. She also discusses donating historical materials, including research equipment, to the Science History Institute. Throughout the interview, Glusker mentions her friendships and professional relationships with other scientists, including Dorothy Hodgkin, Dorothy Semenow, Linus Pauling, Lindo Patterson, Gabrielle "Gai" Donnay, Max Perutz, and others.

Glusker reflects on her achievements and the honors that she has received, sharing that she loved the work that she did throughout her career.

INTERVIEWERS

Jacqueline Boytim is a museum professional based in Philadelphia. She worked at the Science History Institute from 2011-2023, in roles with the Center for Oral History, the museum, the Institute for Research, and Public Engagement. She studied Science, Technology, and Society at the University of Pennsylvania.

Sarah Schneider is a Program Associate in the Center for Oral History at the Science History Institute. She has an interest in preserving and sharing immigration stories in the oral history collection. Schneider holds a BA in American Studies from Brandeis University and an MA in History (Public History track) from the University of Central Florida. She serves as a board member of Oral History in the Mid-Atlantic Region (OHMAR) and was on the 2024 conference committee for the Oral History Association (OHA) annual meeting.

ABOUT THIS TRANSCRIPT

Ann Glusker and Mark Glusker, children of Jenny Pickworth Glusker, reviewed the transcript document. Ann Glusker assisted with confirming details and providing relevant contextual information. Dr. Miriam Rossi also assisted with confirming some information.

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INTERVIEWEE: **Jenny Pickworth Glusker**

INTERVIEWERS: **Jacqueline Boytim**
Sarah Schneider

LOCATION: **Interviewee's home**
Huntingdon Valley, Pennsylvania

DATE: **1 November 2022**

[00:00:00]
SCHNEIDER: So today is Tuesday, November 1, 2022. My name is Sarah Schneider, and I am joined by Jacqueline Boytim. We are conducting the first session of an oral history interview with Dr. Jenny Pickworth Glusker at her home in Huntingdon Valley, Pennsylvania. So thank you for being with us today and participating in the oral history. And so to start off—

[00:00:21]
GLUSKER: Should I start out again? A little bit, yes.

[00:00:23]
SCHNEIDER: Yes, if you could start talking about some of your memories from growing up and tell us where you were born and where you grew up.

[00:00:30]
GLUSKER: Well, I was brought up in England, in Birmingham, which is a big industrial city. No real rivers, but a very good canal system, which is, you know, if you're a child, is of great interest to watch. And my father [Frederick Alfred Pickworth] and my mother [Jane Wylie Stocks] were both physicians at a mental hospital on the edge . . . almost in the country, just suburban Birmingham. Birmingham was built so they had circles of grass areas, and then more housing and then more circles of grass area. So it was rather well designed. And the buildings that he worked in were really nice, and they'd have beautiful gardens so the patients could walk there but couldn't get out. Then they met.

[00:01:31]
And my father's father [Alfred Joseph Pickworth] was a Wesley Methodist minister and a general practitioner, and he had . . . what did he have? He had six children [Florence "Flossie," William, Frederick Alfred, Thomas Claude, Eric, and Ralph], one of whom [Ralph] died at birth. The first three were by his first wife [Sarah Hannah Hart], who was really quite an artist. Somehow, several of us have just a little bit of ability to do artwork, which is nice when you

know that it's come from a particular relative. I'm not quite sure what else I can say. Anyway, here's a picture of my father. I don't know if you can turn that around.

[00:02:20]
BOYTIM: And you were telling us, his mother [Sarah Hannah Hart] died while he was quite young?

[00:02:24]
GLUSKER: Yeah, he was one year old, exactly, yes. His father didn't know what to do. So he called his sisters [Charlotte, Caroline, Jessie, and Emily] to come and help him look after the three children. They squabbled like mad. He sent a message—I have this from my cousin—he sent a message down to the village saying he was interviewing for a new wife and he had three children. Could they send people up? So they sent people up.

[00:02:54]
And what he did was, he gave them a cup of tea, very hot tea, in a saucer, and then he waited to see how they managed to arrange to drink it. It was too hot to drink, but did they put a little in the saucer or what did they do? And I'm not sure what this one wife [Mary Household] did, but she did okay, and so he married her and he had three boys [Thomas Claude, Eric, and Ralph] by her. And they lived in Lakenheath, [England], which no one had ever heard of until World War II, when it was a US Air Force base, Lakenheath. When his father had . . . his father had died, a lot of people would come to his house and knock on the door and there would be an American living in the house. And they'd say, "Is this Pickworth's old place?" Now if you go into town, into the village, you will see the big sign: "Pickworth's old place." So he was remembered as someone . . . .

[00:04:05]
So I had three uncles and one aunt from my father's side. But my father was very good at designing things, and the thing that was most important with him was how he dealt with World War II, because we were young children, I mean, the three of us, in 1939. So he ordered an air raid shelter and it couldn't fit into the house. So he just went into the garden and dug under a flowerbed and put an air raid shelter there. So then he built a little playhouse for us in the garden, and he built a telephone, his own version of a telephone, and he had a telephone to the air raid shelter and a telephone to the house. So we would go and be put to sleep in the air raid shelter under the flowerbeds and we would call him when we could hear the German planes because they sounded different.

[00:05:09]
They handled the gasoline in a different way. So while an English plane would go *zzzzzz*, a German plane would go *zzir-zzir-zzir-zzir*, so you could tell whenever you were about to be bombed. We spent a lot of time during the war in that air raid shelter. But then they would leave about three in the morning because they had to get back over the English Channel before

[00:04:12]
GLUSKER: Oh, you're welcome.

[00:04:14]
SCHNEIDER: Yeah, thank you.

[00:04:15]
GLUSKER: I don't know what else I should be saying to you.

[00:04:18]
BOYTIM: Yeah, I think we're finished.

[00:04:19]
GLUSKER: I mean, I love doing that. I love doing this. I loved doing . . . I loved running a lab. I loved all the editing work I had to do, interacting with people and trying to persuade them not to put a paper in, or write it slightly differently. Some people like to write a letter explaining how they were badly treated by almost everybody else. And I'd say, "Why don't you spend your time talking about how exciting what you found is and then do another paper on the bad world, you know?" And they often are very reluctant to do that, but it helps science if they don't mix the two.

[00:05:04]
BOYTIM: Right. Right. Yeah. It sounds like a healthy approach.

[00:05:08]
GLUSKER: Yeah. Yes. Yes.

[00:05:12]
BOYTIM: Well, I'll end the recording.

[END OF AUDIO, FILE 3.2]

[END OF INTERVIEW]

PUBLICATION LIST

INFRARED SPECTROSCOPY

1953 Pickworth, J. and Thompson, H.W. The fundamental vibration-rotation band of deuterium chloride. Proc. Roy. Soc. (London) A:218 37-43.

1954 Pickworth, J. and Thompson, H.W. Vibration bands and rotational constants of methyl chloride. Trans. Farad. Soc. 50:218-226.

1954 Pickworth, J. and Thompson, H.W. Vibration-rotation bands of methyl fluoride. Proc. Roy. Soc. (London) A:222 443-455.

VITAMIN B_{12}

1954 Brink, C.; Hodgkin, D.C.; Lindsey, J.; Pickworth, J.; Robertson, J.H.; and White, J.G. X-ray crystallographic evidence on the structure of vitamin B_{12}. Nature 174:1169-1171.

1955 Hodgkin, D.C.; Pickworth, J.; Robertson, J.H.; Trueblood, K.N.; Prosen, R.J.; and White, J.G. The crystal structure of the hexacarboxylic acid derived from B_{12} and the molecular structure of the vitamin. Nature176:325-328.

1956 Hodgkin, D.C.; Kamper, J.; Mackay, M.; Pickworth, J.; Trueblood, K.N.; and White, J.G. Structure of Vitamin B_{12}. Nature 178:64-66.

1957 Hodgkin, D.C.; Kamper, J.; Lindsey, J.; Mackay M.; Pickworth, J.; Robertson, J.H.; Shoemaker, C.B.; White, J.G.; Prosen, R.J. and Trueblood, K.N. The structure of vitamin B_{12}. I. An outline of the crystallographic investigations of vitamin B_{12}. Proc. Roy. Soc. A:242 228-263.

1957 Glusker, J.P. Some details of the X-ray analysis of a hexacarboxylic acid obtained from vitamin B_{12}. Acta Cryst. 10:810-811.

1959 Hodgkin, D.C.; Pickworth, J.; Robertson, J.H.; Prosen, R.J.; Sparks, R.A. and Trueblood, K.N. The structure of vitamin B_{12}. II. The crystal structure of a hexacarboxylic acid obtained by the degradation of vitamin B_{12}. Proc. Roy. Soc. A:251:306-352.

1981 Glusker, J.P. Vitamin B_{12}: Introduction and Vitamin B_{12}: Then and Now in "Structures of Molecules of Biological Interest" (Eds. G.G. Dodson, J.P. Glusker and D. Sayre), Oxford University Press, Oxford, England, Ch. 10, pp. 67-71 and Ch. 14, pp. 106-122.

1982 Glusker, J.P. X-ray crystallography of B_{12} and cobaloximes. In "Vitamin B_{12}," Volume 1 (Ed. D. Dolphin), Chapter 3, pp. 23-106. Wiley Interscience: New York.

1985 Rossi, M.; Glusker, J.P.; Randaccio, L.; Summers, M.F.; Toscano, P.J. and Marzilli, L.G. The structure of a B_{12} coenzyme: methylcobalamin studies by X-ray and NMR methods. J. Amer. Chem. Soc. 107:1729-1738.

1986 Marzilli, L.G.; Parker, W.O. Jr.; Kohli, R.K.; Carrell, H.L. and Glusker, J.P. The principal components of synthetic vitamin B_{12}¢. Inorganic Chemistry 25:127-129.

1987 Pett, V.; Liebman, M.N.; Murray-Rust, P.; Prasad, K. and Glusker, J.P. Conformational variability of corrins: some methods of analysis. J. Amer. Chem. Soc. 109:3207-3215.

1987 Pett, V.B.; Jones, R.W.; Mulichak, A.M.; Choo, P.L.; Bacon, J.W.; Zacharias, D.E. and Glusker, J.P. Organocobalt complexes with a buckled equatorial ligand. Acta Cryst. A Abstract 09.4-15.

1988 Rossi, M. and Glusker, J.P. Vitamin B_{12} and its coenzymes: structures and reactivities. In "Environmental Influences and Recognition in Enzyme Activity" (Eds. J.F. Liebman and A. Greenberg), Chapter 1, pp. 1-58. VCH Publishers, Inc.: New York.

1991 Pagano, T.G.; Marzilli, L.G.; Flocco, M.M.; Tsai, C.; Carrell, H.L. and Glusker, J.P. X-ray crystallographic and two-dimensional NMR investigations of a coenzyme B_{12} analogue with 5¢-deoxyadenosine replaced by 9-(CH2)3-adenine. J. Amer. Chem. Soc. 113:531-542.

1995 Glusker, J.P. Vitamin B_{12} and its coenzymes. Vitamins and and Hormones 50:1-76.

1995 George, P.; Glusker, J.P. and Bock, C.W. An *ab initio* study of protonated radical species that may be involved in the enzyme-coenzyme B_{12} catalyzed dehydration of 1,2 dihydroxy-ethane. J. Amer. Chem. Soc. 117:10131-10132.

1996 Glusker, J.P. Vitamin B_{12} and Dorothy. *In* The Collected Works of Dorothy Hodgkin, Volume II, Cholesterol, Penicillin and Other Antibiotics and Vitamin B_{12}, edited by G.G. Dodson, J.P. Glusker, S. Ramaseshan and K. Venkatesan. Interline Publishing: Bangalore, India, 1996, pp. xxxii-xliv.

1997 George, P.; Glusker, J.P. and Bock, C.W. *Ab initio* computational molecular orbital study of radical, protonated radical (radical cation) and carbon cation species that have been proposed in mechanisms for the transfer process in the enzyme-coenzyme B_{12}-catalyzed dehydration of 1,2-dihydroxyethane. J. Amer. Chem. Soc. 119:7065-7074.

1997 Glusker, J.P.; George, P. and Bock, C.W. An *ab initio* molecular orbital study of species that have been proposed in mechanisms for the transfer process in the enzyme-coenzyme B_{12} catalyzed dehydration of 1,2-dihydroxyethane. Biofactors 5:219-221.

1997 Glusker, J.P. Vitamin B_{12}. In Principles of Medical Biology, Volume 8A, Molecular and Cellular Pharmacology (E. Bittar and N. Bittar, eds.). JAI Press Inc.: Greenwich, CT, pp. 909-930 (1997).

1999 George, P.; Siegbahn, P.E.M.; Glusker, J.P. and Bock, C.W. The dehydration step in the enzyme-coenzyme B_{12} catalyzed diol dehydrase reaction of 1,2-dihydroxyethane utilizing a hydrogen-bonded carboxylic acid group as an additional co-factor. A computational study. J. Phys. Chem. 103:7531-7541.

1999 Wagner, T.; Afshar, C.E.; Carrell, H.L.; Glusker, J.P.; Englert, U. and Hogenkamp, H.P.C. Difluoromethylcobalamin: structural aspects of an old tree with a new branch. Inorg. Chem. 38:1785-1794.

2006 Glusker, J.P. Dorothy Crowfoot Hodgkin (1910-1994). *In* Out of the Shadows: Contributions of Twentieth-Century Women to Physics, edited by N. Byers and G. Williams. Cambridge University Press, Chapter 22, pages 240-261.

CALTECH PEPTIDES

1956 Degeilh, R. and Pickworth, J. Unit-cell dimensions and space groups of some simple peptides. Acta Cryst. 9:828.

1961 Marsh, R.E. and Glusker, J.P. The crystal structure of glycylphenylalanyl-glycine. Acta Cryst. 14:1110-1116.

CITRATES

1958 Glusker, J.P., Patterson, A.L.; Love, W.E. and Dornberg, M.L. Crystallographic evidence for the relative configuration of naturally occurring isocitric acid. J. Amer. Chem. Soc. 80:4426-4427.

1960 Glusker, J.P.; van der Helm, D.; Love, W.E.; Dornberg, M.L. and Patterson, A.L. The state of ionisation of crystalline sodium dihydrogen citrate. J. Amer. Chem. Soc. 82:2964-2965.

1963 Glusker, J.P.; Patterson, A.L.; Love, W.E. and Dornberg, M.L. X-ray crystal analysis of the substrates of aconitase. IV. The configuration of the naturally occurring isocitric acid as determined from potassium and rubidium salts of its lactone. Acta Cryst. 16:1102-1107.

1965 Glusker, J.P.; van der Helm, D.; Love, W.E.; Dornberg, M.L.; Minkin, J.A.; Johnson, C.K. and Patterson, A.L. X-ray crystal analysis of the substrates of aconitase. VI. The structures of sodium and lithium dihydrogen citrates. Acta Cryst. 19:561-572.

1967 Gabe, E.J.; Glusker, J.P.; Minkin, J.A. and Patterson, A.L. X-ray crystal analysis of the substrates of aconitase. VII. The structure of lithium ammonium hydrogen citrate monohydrate. Acta Cryst. 22:366-375.

1968 van der Helm, D.; Glusker, J.P.; Johnson, C.K.; Minkin, J.A.; Burow, N.E. and Patterson, A.L. X-ray crystal analysis of the substrates of aconitase. VIII. The structure and absolute

configuration of potassium dihydrogen isocitrate isolated from Bryophyllum calycinum. Acta Cryst. B24:578-592.

1969 Glusker, J.P.; Minkin, J.A. and Patterson, A.L. X-ray crystal analysis of the substrates of aconitase. IX. A refinement of the structure of anhydrous citric acid. Acta Cryst. B25:1066-1072.

1969 Glusker, J.P.; Minkin, J.A.; Casciato, C.A. and Soule, F.B. Absolute configurations of the naturally occurring hydroxycitric acids. Arch. Biochem. Biophys. 132:573-575.

1970 Carrell, H.L.; Glusker, J.P.; Villafranca, J.J.; Mildvan, A.S.; Dummel, R.J. and Kun, E. Fluorocitrate inhibition of aconitase: Relative configuration of inhibitory isomer by X-rays. Science 170:1412-1414.

1971 Glusker, J.P.; Minkin, J.A. and Casciato, C.A. The structure and absolute configuration of the calcium salt of garcinia acid, the lactone of (-)-hydroxycitric acid. Acta Cryst. B27:1284-1293.

1972 Glusker, J.P.; Orehowsky, Jr., W.; Casciato, C.A. and Carrell, H.L. X-ray crystal analysis of the substrates of aconitase. X. The structure of dipotassium cis-aconitate. Acta Cryst. B28:419-425.

1972 Dargay, J.M.; Berman, H.M.; Carrell, H.L. and Glusker, J.P. The crystal structure of potassium dihydrogen trans-aconitate. Acta Cryst. B28:1533-1539.

1972 Glusker, J.P.; Minkin, J.A. and Soule, F.B. The structure and absolute configuration of the calcium salt of hibiscus acid, the lactone of (+)-allo-hydroxycitric acid. Acta Cryst. B28:2499-2505.

1972 Glusker, J.P. Cis- and trans-aconitates: Their probable conformations in the active sites of enzymes utilizing aconitate as a substrate. Arch. Biochem. Biophys. 151:322-327.

1973 Glusker, J.P. and Carrell, H.L. X-ray crystal analysis of the substrates of aconitase. XI. Manganous citrate decahydrate. J. Molec. Struct. 15:151-159.

1973 Carrell, H.L. and Glusker, J.P. Manganous citrate decahydrate. Acta Cryst. B29:638-640.

1973 Carrell, H.L. and Glusker, J.P. The crystal structure of rubidium ammonium fluorocitrate dihydrate. Acta Cryst. B29:674-682.

1973 Berman, H.M.; Carrell, H.L. and Glusker, J.P. Refinement of the structure of potassium hydrogen isocitrate lactone. Acta Cryst. B29:1163-1165.

1973 Glusker, J.P. and Srere, P.A. Citrate enzyme substrates and inhibitors: depiction of their absolute configurations. Bioorganic Chemistry 2:301-310.

1974 Glusker, J.P.; Carrell, H.L.; Job, R. and Bruice, T.C. Mechanism for chiral recognition of a prochiral center, and for amino acid complexation to a Co(III) tetramine. The crystal structure, absolute configuration and circular dichroism of L(-)436-b2-[(2S:9S)-2,9-diamino-4,7-